To Answer or Not to Answer: A Model-Internal Confidence Estimation Approach

Brandon Cheng, Justin Mehes, Trek Stenger

Introduction

Current Large Language Models (LLMs) often suffer from poor calibration. They frequently exhibit high linguistic confidence regardless of factual accuracy, leading to confident hallucinations. This issue stems, in part, from the LLM training objective, which optimizes for next-token probability rather than factual truth. As a result, models can produce fluent and coherent responses that are not grounded in knowledge, while still appearing highly certain. This misalignment between confidence and correctness poses significant challenges, particularly in high-stakes domains such as healthcare, where users may rely on model outputs without the ability to independently verify them. In such settings, overconfident errors can be more harmful than uncertain or abstained responses. The problem is further compounded as LLMs are increasingly deployed as agents capable of executing real-world actions, where a confidently wrong output can have consequences that are difficult or impossible to reverse. Our project is seeking a method that makes decisions for a model based on its internal confidence signals and downstream utility.

Problem Statement

Existing confidence estimation techniques such as using the raw token confidence have been shown to be poorly calibrated and uninformative for reliable decision-making. We recognized the need for richer confidence signals that better reflect a model's true certainty. In our project, we utilize a new method of estimating the confidence level of a model - acquiring intermediate layer outputs and comparing them to the final output. This method was inspired by the original paper that we are extending from: *MICE for CATs: Model-Internal Confidence Estimation for Calibrating Agents with Tools*. We make a decision policy for the model on whether it should answer the question directly, ask for clarification, or abstain from answering given an input and the model's confidence estimation. At the same time, these decisions are made with downstream utility in mind, where incorrectly answering a question could have significant implications. To evaluate the effectiveness of such confidence-driven decision policy, we also train a model to directly predict whether to abstain/clarify/answer and compare results between both approaches.

Approach

Our approach can be outlined by answering three important research questions: 1) How do we benchmark the process? 2) How do we extract confidence/uncertainty? 3) How do we make decisions given uncertainty estimates?

RQ1: Benchmarks - Since our decision-making model outputs one of three actions (abstain, clarify, answer), we evaluate a model's correctness by combining these three datasets: AmbigQA, AbstentionBench, and TriviaQA. AmbigQA is an open-domain question answering dataset introduced by Min et al. (2020) at EMNLP, which addresses the inherent ambiguity of open-domain questions by requiring systems to find every plausible answer and rewrite the question for each one to resolve the ambiguity. AbstentionBench is a recent benchmark from FAIR at Meta (Kirichenko, Ibrahim, Chaudhuri, and Bell, 2025) for evaluating whether LLMs know when not to answer. The benchmark spans 20 datasets across 6 abstention scenarios, covering questions with unknown answers, underspecification, false premises, subjective interpretations, and outdated information. TriviaQA is a large-scale reading comprehension dataset introduced by Joshi, Choi, Weld, and Zettlemoyer (ACL 2017). It has since become a standard benchmark for LLM evaluation as a factual-recall benchmark.

In our evaluation, if the model is prompted with a question from AmbigQA, its correct action is to clarify; if the question is from AbstentionBench, then it supposed to abstain; and if the model is asked with a TriviaQA question, it is expected to answer the question directly.

RQ 2: Confidence/Uncertainty Estimation - We hypothesize that the confidence estimation method introduced in MICE for CATs can be adapted from its original tool-calling setting to our question answering setting. To this end, we implement Logit Lens, a technique that applies the model's unembedding matrix, normally used only at the final layer, to intermediate-layer hidden states in order to extract their predicted token distributions. Then, we compute the BERTScore, a metric of measuring semantic similarity between two text strings, between each intermediate layer and the final layer of the model. The evolution of the BERTScore will be a primary feature when training the confidence estimation model.

In addition to BERTScore, we also train the model using the raw confidence feature, which is the product of token probabilities across the generated response. Specifically, we use the log confidence score to ensure the data is normalized. With the BERTScores and normalized log confidence score, we are left with 32 features to be utilized in generating the predicted confidence scores. The models we use to generate such confidence scores include the Random Forest and Logistic Regression, both of which were also utilized in MICE for CATs.

RQ 3: Decision Making Given a calibrated confidence score for an LLM-generated response, we investigate three different approaches of decision making, each making different assumptions about how confidence relates to optimal action.

<u>Approach 1 - Theoretical utility thresholds:</u> Following the Minimum Bayes Risk (MBR) framework from MICE for CATs, we extend the paper's binary decision to our three-way action space by defining a utility

function U(action, outcome) over all six (action × correctness) pairs. Given a calibrated confidence p, the expected utility of each action is $p * U(action \mid correct) + (1 - p) * U(action \mid incorrect)$, and the model selects the action that maximizes this quantity. We establish three risk profiles: low, medium, and high. We then compute the best action under each risk profile at 10,001 different confidence levels between 0 and 1, and infer the confidence thresholds - the boundary where the optimal action changes.
Approach 2 - Empirical thresholds via validation data: A key limitation of the theoretical approach is that the thresholds depend on utility values chosen by hand rather than on the data distribution itself. To address this, we replace the utility-derived thresholds with thresholds set empirically. Using a 0/1 misclassification loss and a held-out validation set from our 60/20/20 split, we grid-search the upper threshold a and lower threshold b to minimize expected loss under MBR. The decision rule remains the same as approach 1, confidence above a triggers action answer, between b and a triggers action clarify, below b triggers action abstain.
Approach 3 - Direct action prediction: Both threshold-based approaches assume that abstain, clarify, and answer questions occupy distinct regions along a one-dimensional confidence axis. In another approach, we train a separate Random Forest classifier on the same MICE features but with the action label as the target. This approach discards the middle layer of confidence scoring and instead treats action selection as a multi-class classification problem over model-internal features.

We evaluate each approach on the same held-out test set using classification accuracy stratified by question type. Comparing the three approaches lets us isolate where the binary decision making on tool-calling of MICE for CATs do and do not transfer to the three-way decision on question-answering setting.

Additional Methods We used the open-source Meta-Llama-3-8B-Instruct large language model available on HuggingFace. This worked well for our project as it is a well-known, stable model that does not perform tool calling. We began by running 100 of each question type through the model to achieve a baseline accuracy measurement across the three types of questions.

Throughout our analyses we utilize an LLM judge. Originally specified in the 2025 AbstentionBench paper, this LLM judge serves as an automatic scorer that determines whether the output provided by Llama is considered a direct answer, an ask for clarification, or an abstention from answering. This is used to save time so that manual human scoring is not needed.

Results and Analysis

To evaluate our proposed methods in context to the research questions, we first extracted all 32 features from a subset of 2,500 questions drawn across the three datasets. As an initial baseline assessment, we conducted a preliminary test of the LLM's responses by comparing its predicted action against the expected action given the question type. In these preliminary tests, the Llama model produced the correct action 36.4% of the time (912 out of 2,500 questions). The results showed that performance was highly skewed across categories as it answered 97.4% of the questions from the TriviaQA dataset, abstained 11.3% of the Abstainbench questions, and clarified only 0.8% of the AmbigQA dataset questions.

Using the extracted feature set, we constructed a 60/20/20 train/validation/test split. This resulted in 1,500 questions for training the classifier models and 500 questions reserved for the evaluations. With this experimental setup in place, we designed three experiments to assess performance of our proposed methods on the basis of calibration, utility, and benchmark accuracy.

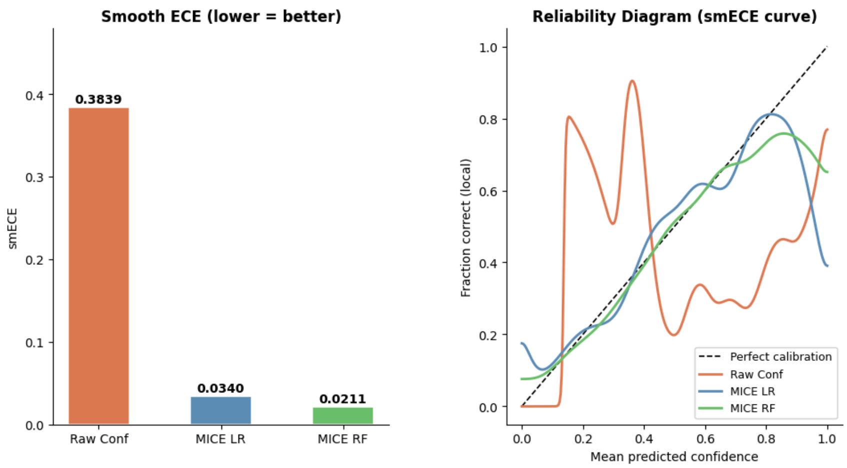

Figure 1: Smooth Expected Calibration Error Results

The first experiment tested the calibration of the raw LLM log-confidence score and our confidence scores produced by the classifier models. The calibration test was conducted by using a variant of the Expected Calibration Error known as the Smooth Expected Calibration Error (smECE). Such variant replaces the inconsistent discrete histogram binning method with the Nadaraya-Watson kernel regression. In our calibration tests, the errors were computed by comparing the confidence estimates against the correctness rates derived from the LLM's baseline performance. The results, shown in Figure 1, indicate that LLMs indeed have poor calibration when just using normalized raw log-probability confidence scores. In contrast, both logistic regression and random forest models achieve substantially lower smECE values, suggesting that supervised classifiers can provide reliable confidence estimates for LLM outputs.

Next, we assessed how each model performed in the three utility profiles - low, medium, and high. The evaluations for utility each apply their thresholds to the held-out test split to obtain the realized utility rates which then are applied to metrics such as ROC AUC and PR AUC. As can be seen in Figure 2, the Random Forest model outperformed the other two models once again across all of the metrics. The performance gap increases as the risk level rises due to the increased penalty when the model is incorrect.

The action clarify was also taken significantly more as the risk increases, while the abstain rate remained the same. This experiment showed that the calibration result from Experiment 1 translates into real downstream utility gains, proving that the model being well calibrated benefits decision making in the utility framework.

	risk	model	ROC_AUC	PR_AUC	avg_utility	answer_rate	clarify_rate	abstain_rate	t_low	t_high
0	low	Raw confidence	0.590767	0.490949	0.0498	0.998	0.000	0.002	0.1999	0.2500
1	low	MICE LR	0.745565	0.616533	0.0774	0.622	0.176	0.202	0.1999	0.2500
2	low	MICE RF	0.764062	0.626465	0.1064	0.632	0.124	0.244	0.1999	0.2500
3	medium	Raw confidence	0.590767	0.490949	-0.2726	0.972	0.026	0.002	0.1999	0.4706
4	medium	MICE LR	0.745565	0.616533	0.0008	0.268	0.530	0.202	0.1999	0.4706
5	medium	MICE RF	0.764062	0.626465	0.0128	0.342	0.414	0.244	0.1999	0.4706
6	high	Raw confidence	0.590767	0.490949	-0.3266	0.082	0.916	0.002	0.1999	0.9073
7	high	MICE LR	0.745565	0.616533	-0.1174	0.006	0.792	0.202	0.1999	0.9073
8	high	MICE RF	0.764062	0.626465	-0.0808	0.000	0.756	0.244	0.1999	0.9073

Figure 2: Utility Summary Table Across Risk Profiles and Models

Both the calibration and utility analysis supported Random Forest as the most reliable model for confidence estimation. Using this specific model and the theoretical utility thresholds of different risk levels, we proceeded to evaluate the decision making of such a system through accuracy tests on our three datasets. As seen below in Figure 3, the medium risk level held the best accuracy. Despite the consistent accuracy results across the different datasets under medium risk level, the theoretical utility thresholds approach for decision making showed to be highly unreliable in the evaluations as the best performing risk level had the overall accuracy score of just 52.2%.

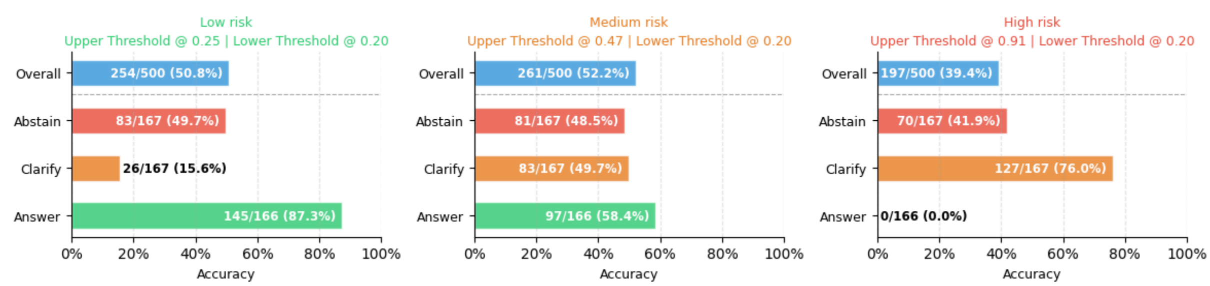

Figure 3: Benchmark Accuracy of Theoretical Utility Threshold Approach in Different Risk Settings

Extending from the original MICE for CATs implementation, we also experimented with an empirical method of finding the decision-making confidence threshold. We found that using this method, we were able to improve the overall accuracy to 52.6%. This method outperformed the theoretical method, which had a 47.5% average accuracy across three risk profiles.

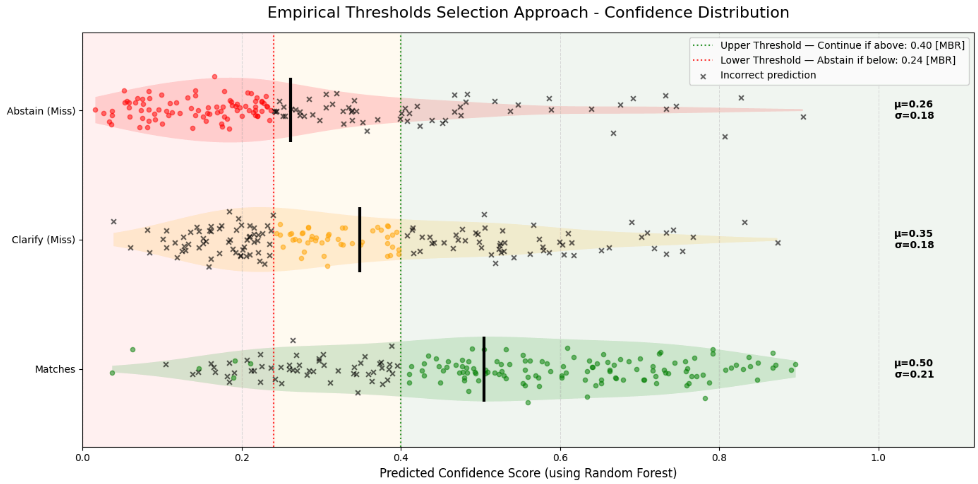

Figure 4: Confidence Score Distribution on the Test Set from the Empirical Method

While the empirical threshold improves the accuracy compared to the theoretical threshold, both accuracy metrics remain subpar. To make sense of this, we examine Figure 4, which shows the confidence score distributions of the test set, with the bottom row being the data points where the baseline LLM responded to questions correctly given their type, the middle row being when the LLM responded incorrectly for questions of type clarify, and the top row being when the LLM responded incorrectly for questions of type abstain. Our hypothesis was that we could find two thresholds on the confidence score distribution separating the three expected action types. However, Figure 4 shows little separation between the three distributions, especially between clarify and abstain, as they are covering roughly the same region on the x-axis. This proved our hypothesis wrong, meaning we cannot find a reliable threshold in the confidence score that distinguishes abstain and clarify.

With this in mind, we attempted another approach hoping to improve the benchmark accuracy. Instead of having the Random Forest output a predicted confidence score, it predicts the action type directly, making it a multi-class classification model. Under this setting, the Random Forest achieved an accuracy of 61%, making it the most accurate method, outperforming the confidence score thresholding approaches.

Conclusion

We found that MICE features calibrate well in the question-answering setting but do not provide a signal strong enough to separate the three action types at decision time. This localizes where the transfer from MICE for CATs breaks: the confidence estimate survives the move from tool-calling to question-answering, but the assumption that abstain, clarify, and answer occupy distinct regions of a one-dimensional confidence axis does not. Future work should test whether this holds across other language models, and should explore additional model-internal signals that could discriminate between clarification and abstention, which raw confidence appears unable to do.